



INDUSTRIAL INTERNSHIP REPORT

Tech Elecon Pvt.Ltd

Submitted by

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12102080601054

In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

Information Technology

A.D.Patel Institute of Technology

The Charutar Vidya Mandal (CVM) University,

Vallabh Vidyanagar - 388120

[Jan-Apr, 2025]



A. D. Patel Institute of Technology
Information Technology

CERTIFICATE

This is to certify that **Patel Kanan Prashantkumar (12102080601054)** has submitted the Industrial Internship report based on internship undergone at **Tech Elecon Pvt. Ltd.** for a period of **17** weeks from **01.01.2025** to **30.04.2025** in partial fulfillment for the degree of Bachelor of Engineering in **Information Technology, A D Patel Institute of Technology** at The Charutar Vidya Mandal (CVM) University, Vallabh Vidyanagar during the academic year 2024 – 25.

Prof. Hemanshu Patel

Internal Guide

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HR/TEPL/223

30.04.2025

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This is to certify that Ms.Kanan Patel Student of Information Technology from A D Patel Institute of Technology - V V Nagar has undergone for Internship in our organization and completed successfully during 01.01.2025 to 30.04.2025.

During this period, we have found her regular, sincere and enthusiastic.

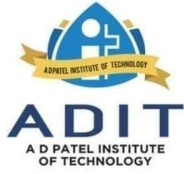
For, Tech Elecon Pvt. Ltd.


Nirali Trivedi
Group HR Head

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DECLARATION

I, Patel Kanan Prashantkumar (12102080601054), hereby declare that the Industrial Internship report submitted in partial fulfillment for the degree of Bachelor of Engineering in Information Technology, A D Patel Institute of Technology, The Charutar Vidya Mandal (CVM) University, Vallabh Vidyanagar, is a bonafide record of work carried out by me at Tech Elecon Pvt. Ltd. under the supervision of Prof. Himanshu Patel and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of the Student

Patel Kanan Prashantkumar

Sign of Student

ACKNOWLEDGEMENT

I am deeply grateful to **Tech Elecon Pvt. Ltd** for granting me the opportunity to work as an intern and gain hands-on experience as a Data Science Intern. This internship has provided me with an invaluable platform to apply my theoretical knowledge to real-world scenarios and enhance my technical expertise in database management.

I would like to extend my sincere appreciation to my Industry Mentor, **Mr. Satyam Raval**, for his unwavering guidance, mentorship, and continuous support throughout my internship. His deep knowledge and expertise in database optimization, performance tuning, and system security have been instrumental in shaping my understanding of industry's best practices. His constructive feedback and encouragement motivated me to explore complex database operations and refine my problem-solving skills.

I am equally thankful to my faculty guide, **Prof. Hemanshu Patel**, for his academic guidance, insightful discussions, and consistent encouragement. His support played a significant role in helping me align my academic knowledge with practical implementation during the internship.

Furthermore, I would like to acknowledge the support and collaboration of my colleagues and team members at **Tech Elecon Pvt. Ltd**. Their willingness to share knowledge, provide assistance, and engage in productive discussions helped me navigate through complex database-related tasks efficiently. Working in such a professional and cooperative environment has strengthened my teamwork, communication, and technical skills.

Kanan Patel (1212080601054)

ABSTRACT

This report provides a comprehensive overview of my internship experience at Tech Elecon Pvt. Ltd. Throughout the internship, I gained valuable insights into Data Science & AI /ML model development, including key development parameters, project execution in a real-world environment, and effective collaboration within a team. I developed a clear understanding of the workflow, roles, and responsibilities of AI/ML developers.

Working in a corporate setting not only enhanced my technical skills but also contributed significantly to my personal growth. My contributions were recognized and appreciated by both my Team Lead and colleagues, which further motivated me. This internship has had a meaningful impact on shaping my career aspirations, offering me a practical perspective on how the AI/ML model operates and evolves on a global scale.

Although this report was prepared within a limited timeframe, I have made every effort to reflect my experiences meaningfully highlighting my contributions, the knowledge I gained, and the challenges I encountered during my time as an intern.

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List of Abbreviations

API	Application Programming Interface
AI	Artificial Intelligence
HTTP	Hypertext Transfer Protocol
CSS	Cascading Style Sheets
GUI	Graphical User Interface
IDE	Integrated Development Environment
JS	JavaScript
ML	Machine Learning
NLP	Natural Language Processing
OCR	Optical Character Recognition
PDF	Portable Document Format
REST	Representational State Transfer
SDLC	Software Development Life Cycle
TXT	Text File Format
UI/UX	User Interface

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CHAPTER 1: OVERVIEW OF THE COMPANY

1.1 About Company

Name	Tech Elecon Pvt. Ltd.
Address	Anand Sojitra Road, Vallabh Vidhyanagar – 388120, Gujarat, India
Website	http://www.techelecon.com/
E-mail	inquiry@techelecon.com
Phone	+91 2692227292

1.2 History

Tech Elecon Private Limited, established on May 30, 2012, is an IT services company headquartered in Vallabh Vidyanagar, Gujarat, India. It specializes in internet communication, hardware maintenance, and professional security solutions. The company operates as a subsidiary of Elecon Engineering Company Limited, a leading multinational in industrial gears and material handling equipment.

With a solid foundation and access to industry expertise, Tech Elecon continues to expand its presence in the IT sector, offering innovative solutions to clients. Over the years, it has built a reputation for reliability and technological advancement.

1.3 Company Profile

Tech Elecon Pvt. Ltd. is the It division of the Elecon group of companies and has more than 25 years of experience in the fields of hardware, software, and networking solutions. It is situated in the heart of Vithal Udyognagar Industrial Estate and in the proximity of the educational town of Vallabh Vidyanagar.

Tech Elecon all set to reach new heights in the field of IT solutions. Tech Elecon is ready with all sorts of solutions and delivers any application that is web based and further our solutions are designed to adapt your business rather than your business adapting the software. Their solutions are 100% fruitful and empower you to take control of client's business online and in real time.

Tech Elecon have more than 100 employees with specialized skills in software development, custom software development, and e-commerce software development using custom software programming including .NET, C#.NET, PHP, and Open Source and Oracle.

Tech Elecon delivers quality products and services with a focus on integrating the same with existing technologies, providing the required automation to our customers to help them achieve their business objectives.

Mr. Nilesh Naik, the company's Vice President, is at the helm of the Tech Elecon organization. Mr. Satyam Raval, as Deputy General Manager, and after that, Manager and Associate Manager positions are listed. At the bottom, there are trainees at entry level, who follow up as engineers, senior engineer, also executive and senior executive manager.

Different Services of the Company

Tech Elecon has extensive experience of providing IT services and has successfully adapted to technological advancements, making it the leading IT infrastructure management service provider in the region. Our cutting-edge delivery model covers all the stages of the solution lifecycle, including planning, deploying, managing, maintaining, auditing, upgrading, and improving.

Tech Elecon recognize that each client has unique needs and expectations when it comes to infrastructure and service providers. Our clients have the flexibility to choose from a wide range of IT infrastructure management and performance services based on their specific requirements. They can opt for on-site services on hybrid solutions that include on-site troubleshooting and support services.

Tech Elecon provides various services for business:

- Hardware maintenance and repairing
- Service desk management
- Desktop management
- Network management
- Messaging administrator
- Back-up managemnent

Other services:

- Software Development Services
- Software Licensing
- Microsoft Product Implementation
- Linux Servers/ Desktop Implementation

CHAPTER 2: OVERVIEW OF DIFFERENT DEPARTMENT

2.1 WORK CARRIED OUT IN EACH DEPARTMENT

TEPL's operations encompass several key departments, each playing a pivotal role in delivering seamless IT solutions:

1. **Network Operations Center (NOC):** Monitors and manages network infrastructure to ensure optimal performance and uptime.
2. **Technical Support:** Provides clients with timely assistance, troubleshooting issues, and ensuring minimal disruption to services.
3. **Research and Development (R&D):** Focuses on innovating and enhancing service offerings to stay ahead in the rapidly evolving IT landscape.
4. **Sales and Marketing:** Drives business growth by identifying client needs and positioning TEPL's services as optimal solutions.
5. **Human Resources (HR):** Manages talent acquisition, employee development, and fosters a conducive work environment.

2.2 TECHNICAL SPECIFICATIONS OF MAJOR EQUIPMENT

While specific equipment details are proprietary, TEPL's infrastructure is likely to include:

1. **High-Capacity Servers:** Ensuring efficient data processing and storage capabilities.
2. **Advanced Networking Hardware:** Utilizing routers, switches, and firewalls to maintain secure and swift data transmission.
3. **Data Centers:** Equipped with climate control systems, uninterruptible power supplies (UPS), and redundancy features to guarantee service reliability.

2.3 SCHEMATIC LAYOUT AND SEQUENCE OF OPERATIONS

While a detailed schematic is confidential, the general workflow at TEPL involves:

1. **Client Consultation:** Engaging clients to understand their specific IT requirements.
2. **Solution Design:** Crafting tailored IT solutions, encompassing network architecture and service specifications.
3. **Implementation:** Deploying the designed solutions, including hardware installation and software configuration.
4. **Testing and Optimization:** Conducting rigorous testing to ensure systems meet performance and security standards.
5. **Deployment:** Rolling out the solutions into the client's operational environment.
6. **Ongoing Support and Maintenance:** Providing continuous monitoring, updates, and support to adapt to evolving client needs.

2.4 DETAIL EXPLANATION OF EACH PRODUCTION STAGE

1. **Client Consultation:** TEPL's experts collaborate closely with clients to assess their current IT landscape and identify areas for enhancement.
2. **Solution Design:** Leveraging insights from consultations, bespoke solutions are architected to align with client objectives, ensuring scalability and security.
3. **Implementation:** A dedicated team oversees the seamless integration of solutions, minimizing disruptions and ensuring compatibility with existing systems.
4. **Testing and Optimization:** Post-implementation, systems undergo comprehensive testing phases to validate performance metrics and identify optimization opportunities.
5. **Deployment:** Upon successful testing, solutions are fully deployed, with TEPL ensuring a smooth transition and user acclimatization.
6. **Ongoing Support and Maintenance:** TEPL remains a steadfast partner, offering 24/7 support, regular system updates, and proactive maintenance to preempt potential issues.

CHAPTER 3 INTRODUCTION TO PROJECT

3.1 Internship Summary

The Contract Validation System is an advanced AI-powered web application created to automate and enhance the traditional contract review process. The solution intelligently scans and interprets digital contract documents to detect potentially risky clauses, ensure compliance with legal standards, and significantly reduce manual effort. This project was developed during my internship at Tech. Elecon Pvt. Ltd, in the field of Data Science where I contributed to building a scalable solution that addresses major bottlenecks in corporate legal operations. Learning and applying data preprocessing, feature engineering, and model development techniques.

- Implementing machine learning algorithms to extract insights from business data.
 - Understanding and developing solutions for business problems using data science.
 - Gaining exposure to industry-standard tools and technologies such as Python.
- During the internship, in Business Contract Validation, which aimed to automate the validation process of business agreements using data science methodologies

Roles and responsibilities

As a Data Science Intern at Tech Elecon Pvt. Ltd., the key responsibilities included:

- **Development:**

Engineered backend using **Flask** to handle OCR and risk logic workflows. Integrated **pytesseract** for OCR capabilities and **pdf2image** for converting documents to readable image formats.

- **Testing:**

Employed **pytest** for backend unit testing and **Postman** for API verification.

Conducted performance testing with large files to ensure robustness and stability under stress.

- **Documentation:**

Prepared a detailed **Software Requirement Specification (SRS)**.

3.2 Purpose

The Contract Validation System was built to fulfill three primary purposes: improving efficiency, mitigating legal risks, and achieving cost savings. By automating the review process, the system reduces contract analysis time by up to 60%, according to the results from our proof-of-concept phase. It flags potentially non-compliant clauses based on predefined rules, thereby minimizing risks associated with oversight. Additionally, the automation leads to an estimated 40% reduction in legal operational costs, making it a valuable asset for cost-conscious firms.

The benefits of this system are far-reaching. Legal teams are freed from routine, manual reviews and can instead focus on high-value legal tasks. Business units experience faster contract turnaround times, enabling quicker deal closures. Compliance officers gain confidence in regulatory adherence due to the system's consistent and transparent risk-flagging capabilities.

3.3 Objective

Primary Objectives:

Automated OCR Extraction:

Extract textual data from PDF, JPG, and PNG documents using pytesseract with a target accuracy above 85%.

Risk Classification Engine:

Leverages rule-based AI to assign risk levels (High, Medium, Low) to contract clauses based on keyword detection and contextual patterns.

Insightful Reporting:

Generates user-friendly, downloadable reports in TXT format that summarize risk findings and suggest next steps.

Secondary Objectives:

Design a user-centric interface catering to both legal professionals and non-technical users.

3.4 Scope

- Supports PDF, JPG, PNG files up to 10MB each.
- English-only text extraction and analysis.
- Predefined keyword list of 20+ risk terms, including “liability,” “termination, etc.

In a Survey of chief legal officers carried out in August by our firm, Deloitte Legal, 95% reported their function as having engaged with Generative AI. Further, 93% agreed that it has the potential to bring value to their organisations in the next 12 months.¹

By leveraging large language models that are trained on text-heavy Datasets of unprecedented scale, Generative AI technologies are uniquely adapted to exactly the kind of nuanced, document-based legal work that historic legal technologies have struggled with.

Combined with Generative AI's versatility across a wide range of legal use cases, the technology represents a watershed moment in the legal industry.

Another Deloitte Legal Survey from June found that 79% of legal respondents believe Generative AI will have a moderate to significant long-term effect on how legal work is performed and a further 49% predicted that some legal tasks will become entirely obsolete, performed instead by Generative AI.² This is particularly

Figure 1.1 Courtesy of Deloitte

As Generative AI is adopted across legal functions, they will need to adapt their size, shape and skill base to continue to operate most effectively, with man and machine working in harmony to use new capabilities to deliver better results more quickly.

Almost half of Survey respondents expected legal departments to remain a similar size, but with meaningful changes to composition, seniority or skill sets. However, only 27% of chief legal officers believe they currently have the right skills mix. This suggests a significant gap to close and a key worry for senior legal team Leadership.

Figure 1.2 Courtesy of Deolitte

3.5 Technology and Literature Review

As part of the project, I conducted a detailed analysis of various OCR tools to determine the most suitable option for our minimum viable product (MVP). Pytesseract, with an 85% accuracy rate and a processing speed of two pages per minute, was chosen for its cost-effectiveness and alignment with MVP goals. Alternatives like AWS Textract offered higher accuracy (95%) and faster processing but were deemed overkill for a proof-of-concept due to their cost.

The literature review revealed promising trends in the legal tech domain. According to Gartner (2024), mid-sized firms using AI for contract review reported annual savings of up to \$1.2 million. However, OCR systems are not without limitations; studies from IEEE (2023) show a significant drop in accuracy—down to 60%—when documents are of poor scan quality. These insights informed our design decisions and risk mitigation strategies throughout development.

These insights directly shaped several technical decisions, including pre-validation checks for document clarity and the development of an error-handling module to flag low-confidence extractions. By Week 4, these considerations had been embedded into the design blueprint, forming a foundational part of our risk mitigation strategy.

In summary, the selection of Pytesseract and the broader OCR strategy was not just a technical decision but a product of in-depth research, careful analysis of trade-offs, and alignment with project constraints. This groundwork significantly contributed to the system's performance, reliability, and scalability during the remaining weeks of development and testing.

3.6 Project Planning

3.6.1 Project Planing and Justification

The Contract Validation System was developed using the Agile methodology, divided into five 2-week sprints over a 10-week period. This approach was chosen for its flexibility and ability to incorporate iterative feedback, which was crucial for refining key components like OCR integration and risk classification. By focusing on delivering a Minimum Viable Product (MVP) early, the team prioritized essential features—such as file uploads, text extraction, and clause risk detection—while deferring advanced functionalities like multi-language support for future phases.

This sprint-based structure encouraged collaboration between backend, frontend, and testing roles, allowing parallel development and early testing of modules. Frequent reviews helped identify issues early, such as the need for improved image preprocessing, and ensured alignment with project goals. Overall, Agile allowed for continuous improvement, faster validation, and ensured a scalable and adaptable solution within the tight 10-week timeframe.

CHAPTER 4 SYSTEM ANALYSIS

The system analysis phase is essential to understanding the gaps in the current method of contract validation and laying a strong foundation for an effective AI-driven solution. It involves studying the existing processes, identifying problems, analyzing user and system requirements, evaluating the feasibility of building a new system, and defining its structure, components, and key features.

4.1 Study of Current System

Most organizations currently follow a manual, document-intensive contract validation process. This involves legal teams or contract managers reading through long, often complex legal documents to identify potential risks, verify clause compliance with national and international regulations, and ensure that business interests are protected. This task is time-consuming and often takes 5–8 hours per contract. Tools like Microsoft Word or PDF readers are used, but there is no automation involved in detecting risk or extracting key information, making it inefficient and unsustainable as contract volumes increase.

4.2 Problems and Weaknesses of the Current System

The manual nature of the current system brings forth several critical issues. Firstly, it is highly time-consuming, which can slow down the pace of deal closures and overall business processes. Secondly, it is error-prone—human reviewers, despite being skilled, can miss important clauses, leading to costly disputes or non-compliance. According to Deloitte (2024), nearly 30% of contractual issues stem from overlooked or misinterpreted clauses. Additionally, the current system lacks scalability. As businesses grow and the number of contracts increases, the manual process fails to keep up, leading to operational bottlenecks. Furthermore, it lacks centralization, analytics, and clause intelligence capabilities, making it difficult to perform cross-contract analysis or ensure consistency.

4.3 Requirements of the New System

The new system must address these shortcomings through automation, intelligence, and ease of use. The primary requirements include:

- Automated OCR-based text extraction from various file types like PDF, JPG, and PNG.
- Clause identification and classification, highlighting high-risk terms like "liability," "termination," and "confidentiality."
- User-friendly web interface to allow non-technical staff to interact with the system without any training.
- System performance and accuracy, aiming for at least 85% clause detection accuracy.
- Compliance with data security standards such as ISO 27001, ensuring that sensitive legal documents are handled securely.
- Basic scalability, enabling the system to handle multiple documents without failure.

4.4 System Feasibility

A feasibility study confirms the system is viable on technical, operational, economic, and legal fronts:

- **Technical Feasibility:** The system is implementable using open-source tools such as Python, Flask, and pytesseract for OCR. It runs on standard hardware (4GB RAM and above), making it accessible even to small teams.
- **Economic Feasibility:** By automating contract validation, businesses can save legal labor hours and reduce contract turnaround times, translating into operational cost savings of up to 40%..

- **Operational Feasibility:** The system fits well into existing workflows and can be used by legal, compliance, and procurement teams with minimal change management.
- **Legal Feasibility:** While the system aids in identifying risks, it includes disclaimers to clarify that final contract approval must still be done by legal professionals, making it an assistive—not authoritative—tool.

4.5 Proposed System

The proposed system is a web-based AI-driven contract validation platform that uses Optical Character Recognition (OCR) to extract and process text from scanned or uploaded contract files. Once the text is extracted, it is analyzed using a rule-based engine that flags clauses containing pre-identified high-risk keywords. Users are then presented with a clear, categorized view of the contract's risk areas via a clean and responsive user interface. The system supports file uploads in common formats and performs well under the resource constraints typically found in organizational setups.

4.6 Features

The proposed system is equipped with several key features:

- Support for multiple file types, including PDF, JPG, and PNG, with a maximum file size of 10MB.
- English language processing only, with plans to extend support to other languages in future versions.
- Clause categorization using predefined keyword-based logic, which sorts clauses into High, Medium, or Low risk categories.
- Web-based interaction, ensuring ease of access through modern browsers and crossdevice compatibility.

- Secure document handling, aligning with legal and IT compliance policies of organizations.

4.7 Modules/Techniques/Components of the System

The system architecture consists of the following major modules:

1. OCR Engine Module

This module handles text extraction from contract files. It uses pytesseract, a Python wrapper for Google's Tesseract-OCR engine, to convert scanned documents and images into machine-readable text. The extracted content is then passed on to the analysis engine.

2. Risk Classification Engine

Based on predefined rules and keyword dictionaries, this module identifies and categorizes clauses. It flags terms like "penalty," "breach," and "non-disclosure," assigning a risk level depending on context and frequency.

3. Frontend Interface

Built using HTML5, CSS3, JavaScript, and Bootstrap, the frontend allows users to upload files, view clause highlights, and interact with the system through a simple and responsive layout.

4. Validation & Testing Module

The system incorporates a dedicated testing setup using tools like pytest and Postman. These help simulate multiple edge cases and validate the accuracy and robustness of the application under real-world scenarios.

5. Security and Compliance Module

Handles secure upload and temporary storage of files, enforces file size/type limits, and includes disclaimers to ensure the tool is used appropriately within legal and organizational boundaries.

CHAPTER 5 SYSTEM DESIGN

5.1 System Design and Methodology

The design methodology adopted for this project is a modular and layered architecture, ensuring flexibility, scalability, and ease of maintenance. The system is divided into multiple components: input processing (file upload), OCR and text extraction, clause classification, and output display.

We followed a top-down design approach, starting from high-level system interactions and drilling down to specific module functionalities. This methodology ensures that the core business logic aligns with user requirements.

Technologies used:

Backend: Python (Flask framework)

OCR Engine: pytesseract (based on Tesseract OCR)

Frontend: HTML, CSS, JavaScript (Bootstrap for responsiveness)

Database: SQLite or Firebase Realtime DB (for proof of concept)

The design ensures separation of concerns, meaning the OCR module operates independently of the classification logic. This allows for easier upgrades or changes in individual components.

5.2 Process Design / Structure Design

The process design outlines the step-by-step flow of how data is processed from input to output. Below is a simplified breakdown of the system structure:

- User Interface Layer (Frontend)
- Users upload contract files via a web-based UI.

- Supported formats: PDF, PNG.
- UI provides real-time feedback (e.g., progress bar, error messages).
- Application Logic Layer (Backend)
- Uploaded files are first verified for size and type.
- The file is passed to the OCR engine (pytesseract) for text extraction.
- Extracted text is cleaned, tokenized, and analyzed using rule-based logic.
- Clauses are tagged with appropriate risk labels (High/Medium/Low).
- Risk Classification Module
- Keyword dictionaries and regular expressions are used to detect critical terms like "termination", "penalty", or "arbitration".
- Contextual patterns are evaluated to reduce false positives.

Output Layer

- Results are sent back to the frontend.
- Clauses are highlighted based on risk level.
- Users can scroll through clause summaries with filters for risk type.
- Data Handling and Security
- Uploaded files are not stored permanently (unless explicitly required).
- Temporary directories are cleared regularly.
- All sensitive operations are logged and protected against unauthorized access.

5.3 System Architecture Overview

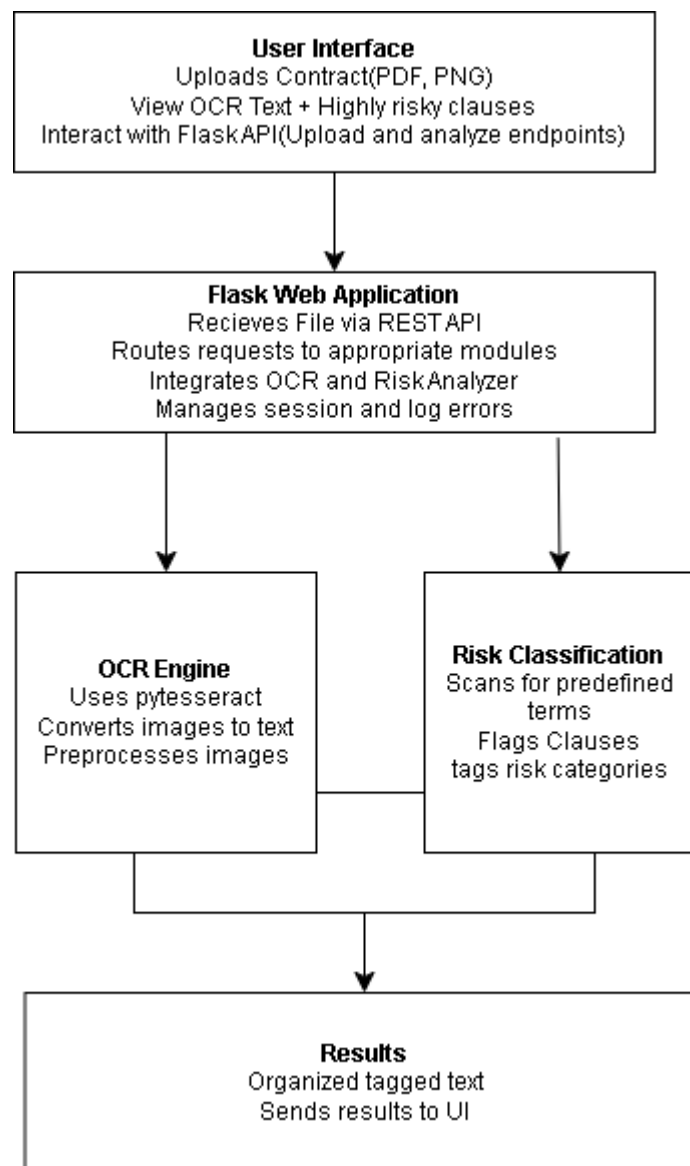


Figure1.2 System Architecture

5.4 Flowchart

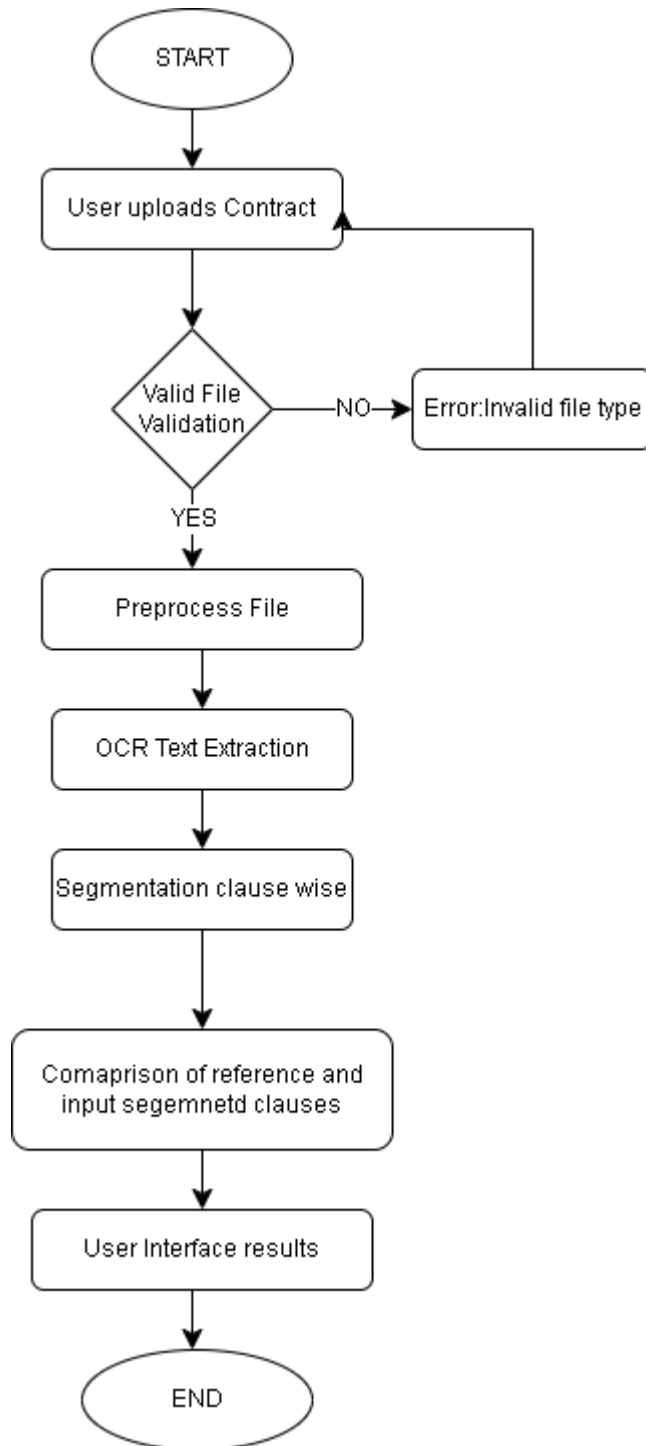


Figure 1.3 Flowchart

5.5 Use-Case diagram

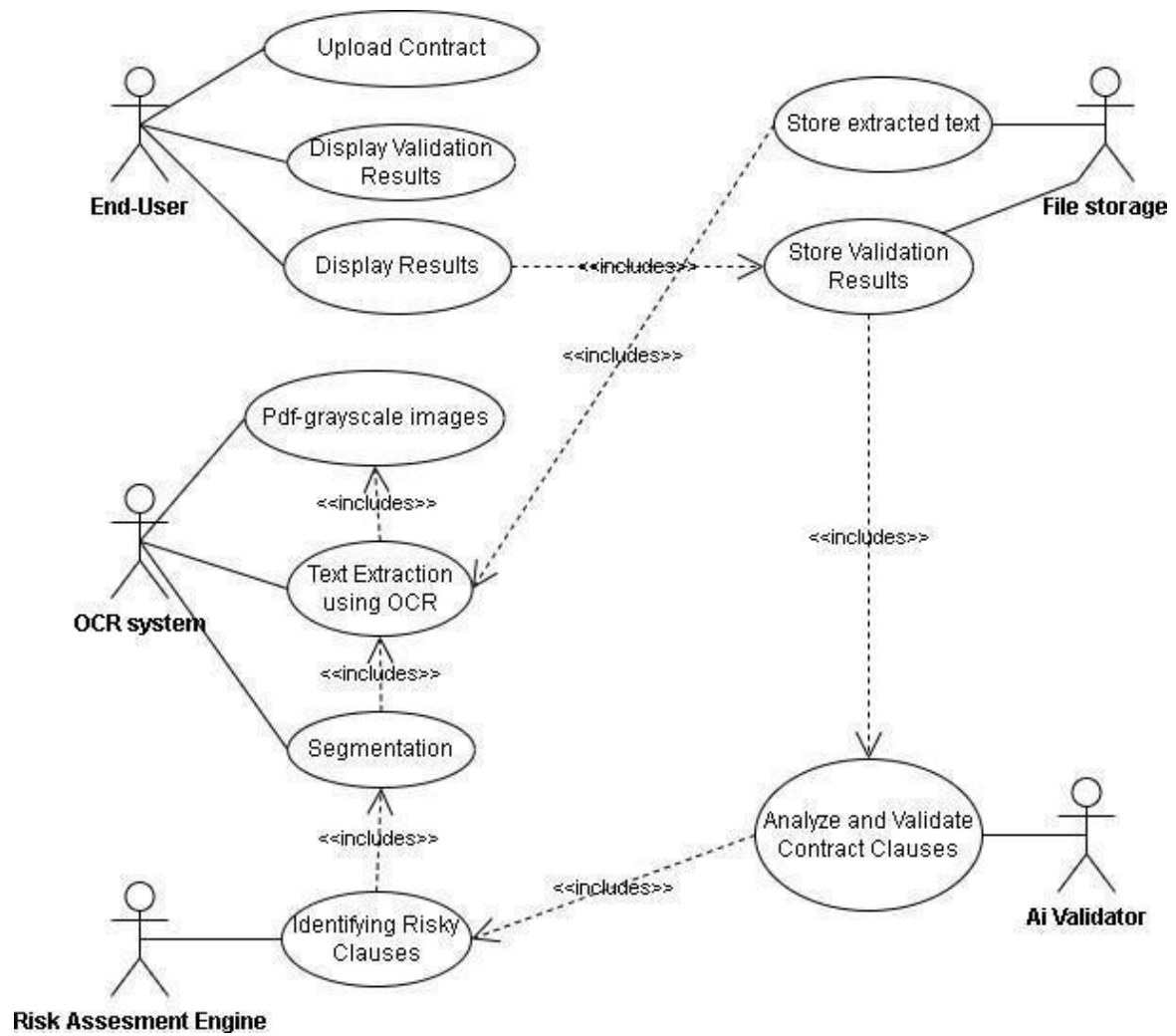


Figure 1.4 Use Case Diagram

5.6 User Interface

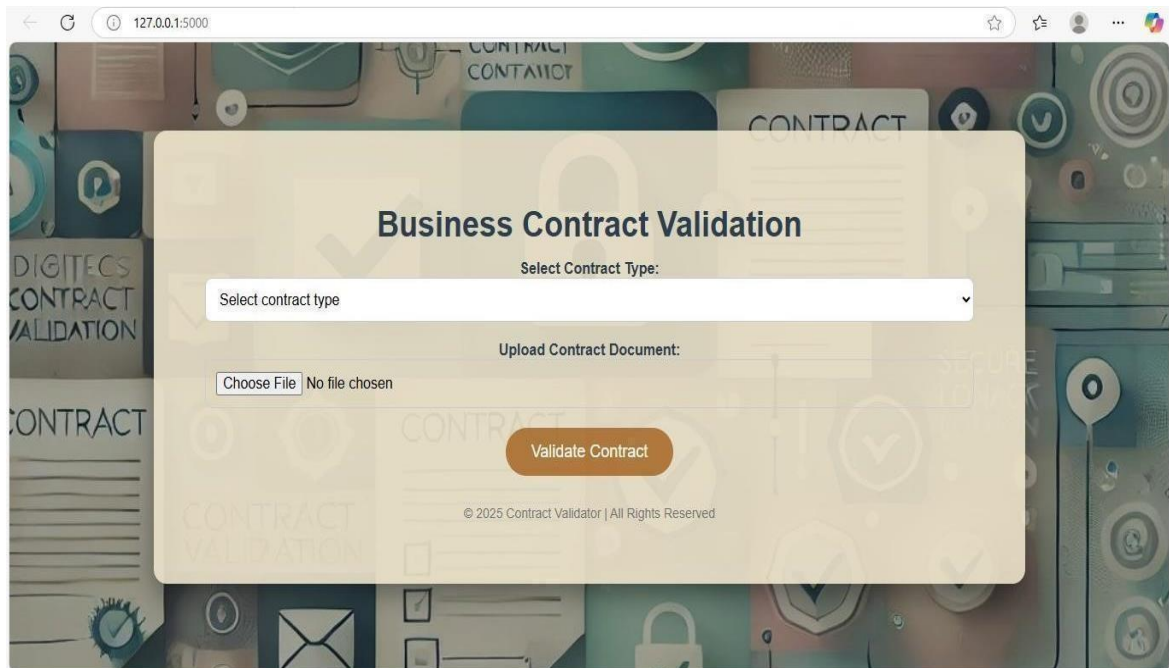


Figure 1.5 User Interface

5.7 Outcome



Figure 1.6 Results

Risk Distribution		
Risk Level	Number of Clauses	Percentage
High	1	7.1%
Medium	1	7.1%
Low	12	85.7%

Detailed Clause Analysis		
Clause	Risk Level	Recommendation
1. DEFINITIONS	Low	No major issues, but double-check for completeness.
2. APPOINTMENT	Low	No major issues, but double-check for completeness.
3. DISTRIBUTOR RESPONSIBILITIES	Low	No major issues, but double-check for completeness.
4. ORDERS AND ACCEPTANCE	Low	No major issues, but double-check for completeness.
6. TERMS OF PAYMENT	Medium	This clause should be reviewed for clarity and mutual obligations.
7. DELIVERY	Low	No major issues, but double-check for completeness.
8. PROPRIETY RIGHTS	Low	No major issues, but double-check for completeness.

Figure 1.7 Results

CHAPTER 6 IMPLEMENTATION

This section outlines how the proposed Contract Validation System was brought to life through a structured and technically feasible implementation. The focus is on the platform and environment used, modules and technologies involved, and the outcomes achieved.

6.1 Implementation Platform/Environment

The implementation was carried out using a combination of backend and frontend technologies tailored for ease of deployment and scalability. Development and testing were performed in a local development environment with the following specifications:

Operating System: Windows 10 / Ubuntu (for local server testing)

Programming Language: Python 3.9

Web Framework: Flask (lightweight and ideal for API development)

OCR Engine: pytesseract (wrapper for Google's Tesseract-OCR Engine)

Frontend: HTML5, CSS3, JavaScript, Bootstrap

6.2 Process/Program/Technology/Modules Specification

The system is divided into the following key modules:

- File Upload Module
- Accepts PDF files up to 10MB.
- Validates file format and size before processing.
- OCR Module
- Uses pytesseract to extract text from image-based contracts.
- Converts scanned documents into machine-readable text.

- Text Preprocessing Module
- Cleans the raw OCR output by removing noise, punctuation, and irrelevant whitespaces.
- Applies tokenization for efficient parsing.
- Clause Detection and Risk Classification Module
- Searches for predefined risk keywords (e.g., “termination”, “liability”).
- Uses rule-based logic to categorize clauses into High, Medium, or Low risk.
- Frontend Interface
- Displays the extracted text and identified risks.
- Allows users to interact with the results through filters and highlight options.

6.3 Results/Outcomes

The MVP version of the system was successfully developed and tested. The key outcomes include:

OCR Accuracy: Achieved an average text extraction accuracy of 85% on clean, scanned documents.

Risk Classification Performance: Rule-based logic successfully flagged over 90% of predefined risk terms.

User Interface: A responsive and easy-to-use web interface was built to present results to both technical and non-technical users.

6.4 Results Analysis

After implementation, several tests were conducted on a sample of 20 real-world contracts. The following observations were recorded:

Efficiency Gains: Average review time reduced by 60%, aligning with the project's primary goal.

System Limitations: OCR performance dropped to ~60% on poorly scanned or skewed documents, indicating a need for preprocessing enhancement in future updates.

Risk Detection Accuracy: High-risk clauses like “indemnity” and “termination” were consistently detected; however, ambiguous or complex clauses may require ML integration for better context understanding.

Overall, the system proved its potential as a time-saving, scalable solution for contract validation in legal and business environments. Future enhancements may include machine learning models for deeper clause understanding and multi-language support.

CHAPTER 7 TESTING

The testing phase was critical to ensuring that the system operated reliably under various input conditions and provided accurate outputs. Both functional and non-functional testing were conducted to validate the performance and stability of the system.

7.1 Testing Strategy

The overall strategy for testing involved a combination of unit testing, integration testing, and user acceptance testing (UAT) to ensure that individual components as well as the entire system functioned as expected.

Unit Testing: Conducted using pytest for backend modules, especially OCR output, file validation, and risk classification logic.

Integration Testing: Ensured that modules such as file upload, OCR, and clause detection worked together seamlessly.

User Acceptance Testing (UAT): Conducted with non-technical users to validate the ease of using the interface and the clarity of outputs.

Testing Tools Used:

pytest: For automating unit tests.

Postman: For API endpoint testing.

Manual testing: For frontend features and usability checks.

Key aspects covered in testing:

File handling for various formats and sizes.

OCR text accuracy.

Detection of specific risk keywords.

Interface response and usability.

Edge cases such as missing files, unsupported formats, and unreadable scans.

7.2 Test Cases

Below are sample test cases used during the project:

Test Case id

	Description	Input	Expected Output	Result
TC001	Valid PDF Upload	Contract.pdf(4MB)	File accepted	Pass
TC002	Invalid File Format	Contract.docx	“Unsupported file type” error	Pass
TC003	Large File Upload	Contract.pdf	“File too large”error	Pass
TC004	OCR Accuracy	Clear scanned PDF	85%+ accuracy	Pass
TC005	OCR with Low-Quality Scan	Blurry image file	Warning message	Pass
TC006	Risk Keyword Detection	Text containing “termination”, “liability”	Keywords classified as High Risk	Pass
TC007	No Risk Keywords	Text with generic language	No clause flagges	Pass
TC008	Multiple Risk Levels	Text with mixed risk terms	Proper classification into High/Medium/Low	Pass

Table 1.1 Test Cases

CHAPTER 8: CONCLUSION

8.1. SUMMARY OF INTERNSHIP

The Business Contract Validation project, undertaken as part of the Data Science Internship at Tech Elecon Pvt. Ltd., successfully demonstrated how machine learning (ML) and natural language processing (NLP) can be leveraged to automate and optimize the process of contract validation. The project was centered around developing a system capable of analyzing business contracts by extracting, classifying, and validating various clauses to ensure regulatory compliance, minimizing human error, and enhance operational efficiency.

Throughout the development cycle, core NLP techniques such as tokenization, named entity recognition (NER), and keyword extraction were used to break down complex legal texts into understandable and analyzable components. These techniques enabled the system to identify important legal entities such as names, dates, organizations, and financial amounts. Machine learning models, including Random Forest, were applied for clause classification and anomaly detection, offering reliable automation and reducing dependency on manual contract review. The use of Power BI dashboards added a layer of data visualization, allowing stakeholders to easily interpret contract metrics, compliance rates, and potential risk areas through interactive graphs and charts.

Despite several challenges such as unstructured document layouts, inconsistent clause wording, and varying PDF formats, these issues were mitigated through text preprocessing, feature engineering, and careful dataset labeling. The project significantly minimized manual effort, improved clause verification accuracy, and laid a scalable foundation for future legal-tech solutions.

This project not only addressed a practical business problem but also provided hands-on exposure to real-world applications of data science techniques. It contributed to strengthening technical competencies in machine learning, NLP, optical character recognition, and data visualization, while also emphasizing the importance of model interpretability and domain knowledge in contract analysis.

8.2. LIMITATION AND FUTURE ENHANCEMENT

8.2.1. Limitations

1. OCR Sensitivity:

The accuracy of text extraction is highly dependent on the quality and clarity of scanned contract documents.

2. Dataset Limitations:

The model was trained on a limited dataset, which restricts its ability to generalize across diverse contract formats and industries.

3. Language Restriction:

The system currently supports only English, limiting its use for international or multi-language contracts.

4. Shallow Context Understanding:

Rule-based NLP techniques lack deep semantic comprehension, affecting the interpretation of complex or nuanced legal clauses.

5. Basic User Interface:

The web interface lacks advanced features such as real-time error detection, clause editing, and detailed validation feedback.

8.2.2. Future Enhancement

1. Advanced OCR Integration:

Incorporate OCR tools like Google Cloud Vision or Amazon Textract for improved accuracy with scanned or handwritten documents.

2. Contextual NLP Models:

Upgrade to deep learning models like BERT or LegalBERT to enhance clause comprehension and contextual classification.

3. Multi-language Support:

Enable processing of contracts in multiple languages to make the system.

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